

Author biography

Manuscript title: Predictive fidelity versus training utility in surrogate-based reinforcement learning for HVAC control: a role-separated hybrid remedy

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Almaz Sapargali received the B.Sc. degree in cyberphysical systems from the International Information Technology University (IITU), Almaty, Kazakhstan, in 2023, and is currently pursuing the M.Sc. degree in computer science and technology at Al-Farabi Kazakh National University, Almaty, Kazakhstan, under a double-degree programme with Northwestern Polytechnical University (NPU), China. Research interests include deep reinforcement learning and scientific machine learning, with a focus on surrogate-based control of building energy systems. Earlier results from the present study were presented at the WCCM–ECCOMAS Congress 2026 (Munich, Germany, 19–24 July 2026) under the title “Deep reinforcement learning for energy-efficient control in smart buildings”. In the present work the contributions covered the methodology, the software, the experiments and the original draft.